	GLOBAL ECONOMICS **Grade 11** **Catch ups exam** **Continuous distributions and sample data** **Teacher: Nicolás López Cuéllar**	**1st TERM** **2026–2027**

Learning objective: Identify probability distributions and their supports; use geometric interval probabilities and exponential waiting times; calculate theoretical moments, empirical probabilities, and sample statistics; and interpret errors relative to theoretical models in economic and operational contexts.	**Assessment criteria:** C1. Characterises discrete and continuous random variables, identifies suitable distributions and supports, and interprets density and cumulative probability in context. C2. Applies uniform and triangular models and constant or piecewise-linear densities to interval probabilities using geometric areas. C3. Uses circular geometry and arc proportions to calculate probabilities for uniform timing and directions around a circle. C4. Applies exponential models to constant-rate waiting times, infers event rates, and assesses model suitability. C5. Calculates and interprets the mean, variance, and standard deviation of continuous distributions using standard formulas. C6. Estimates interval probabilities using sample relative frequencies and compares them with theoretical probabilities. C7. Calculates sample means, variances, and standard deviations for different sample sizes and data representations and compares them with theory.

Name: _____ **Date:** _____

Criteria assessment

This exam assesses C1–C7 for a total of **41 points**. Each question and task shows its points. **Passing requires at least 4 points per criterion**. Show all procedures; answers without the required procedure receive no credit. Partial credit may be awarded for a correct method with arithmetic errors. Missing names or late submissions incur a 1-point deduction per criterion.

Use unrounded intermediate values, including when calculating errors. Round final statistics and absolute errors to 3 decimals, and percentage errors to 2 decimals. Give variance in squared units and standard deviation in the original units. For other probabilities and rates, give an exact expression or at least 3 decimal places. Use geometric areas, not integration, in Questions 2–3.

Question	C1	C2	C3	C4	C5	C6	C7	Question	C1	C2	C3	C4	C5	C6	C7
1	5	–	–	–	–	–	–	12	–	–	–	–	2	–	–
2	–	3	–	–	–	–	–	13	–	–	–	–	2	–	–
3	–	4	–	–	–	–	–	14	–	–	–	–	2	–	–
4	–	–	2	–	–	–	–	15	–	–	–	–	–	1	–
5	–	–	3	–	–	–	–	16	–	–	–	–	–	1	–
6	–	–	–	0.5	–	–	–	17	–	–	–	–	–	2	–
7	–	–	–	0.5	–	–	–	18	–	–	–	–	–	2	–
8	–	–	–	1	–	–	–	19	–	–	–	–	–	–	1
9	–	–	–	1	–	–	–	20	–	–	–	–	–	–	1
10	–	–	–	1	–	–	–	21	–	–	–	–	–	–	2
11	–	–	–	2	–	–	–	22	–	–	–	–	–	–	2
Total	**5**	**7**	**5**	**6**	**6**	**6**	**6**	**Overall total: 41 points**							

1. Models in organisations [C1-5]

Colombian organisations model the following uncertain quantities.

(C1) For each context, define or characterise the random variable and its possible values (support), classify it as discrete or continuous, and identify its distribution. Each context is worth 0.5 points. **[5]**

Context 1. A cooperative records whether one randomly selected member repays a monthly instalment on time. The probability of timely payment is 0.92; code timely payment as 1 and otherwise as 0.

Context 2. A warehouse models a scale's measurement error between -3 and 3 grams, with equal-length intervals having equal probabilities and no errors outside these limits.

Context 3. A bookseller contacts 40 customers. Each independently orders a new title with the same probability 0.15. The bookseller counts the customers who order.

Context 4. A transport cooperative counts roadside assistance calls in one hour. Calls occur independently, one at a time, at a constant rate of 2 per hour; multiple calls in a sufficiently short interval are negligible.

Context 5. A printer models the completion time of a poster order on $[10, 30]$ minutes. The density rises linearly from zero at 10 to a single peak at 18, then falls linearly to zero at 30; it is zero elsewhere.

Context 6. An auditor checks invoices independently until finding the first incorrect one. Each invoice has the same probability 0.08 of being incorrect. Count all invoices checked, including the first incorrect invoice.

Context 7. A cooperative models the fraction of a storage tank used at dispatch on $[0, 1]$. Its density increases along one straight line from zero at 0 to its maximum at 1 and is zero outside the interval.

Context 8. A seed supplier checks 25 independently selected packets. Each packet has the same probability 0.04 of a broken seal. The supplier counts packets with broken seals.

Context 9. A roaster assumes bag mass follows the exact bell-curve model on the real line, with mean 500 grams and standard deviation 4 grams.

Context 10. A digital shop counts refund requests over a three-day period. Requests occur independently at a constant rate of 5 per day, one at a time; multiple requests in a sufficiently short interval are negligible.

2. Dispatch delays [C2-3]

A Medellín supplier models the additional dispatch delay D in minutes on $[0, 12]$. Its density is a single decreasing straight line, zero outside this interval. The density height at 0 is three times the density height at 12. This is a ratio of density heights, not of probabilities at individual points.

(C2) Infer the normalising endpoint heights and use geometric areas to find $P(3 \leq D \leq 9)$. **[3]**

3. Market-stall setup [C2-4]

The setup time S for a stall at a Bogotá craft market ranges from 5 to 25 minutes. Its density is zero at both endpoints, rises linearly to its single maximum at 13 minutes, then falls linearly to zero; it is zero outside $[5, 25]$.

(C2) Use normalisation and geometric areas to find $P(9 \leq S \leq 19)$. **[4]**

4. A repeating inspection cycle [C3-2]

A cold-chain alarm occurs at a uniformly distributed position within a repeating 60-minute inspection cycle. The cycle starts at minute 0 and restarts immediately after minute 60.

(C3) Find the probability that the alarm occurs in the window from minute 52 through the restart to minute 11 of the next cycle. Use arc proportions on the cycle. **[2]**

5. Direction of a survey drone [C3-3]

A survey drone's initial heading is uniformly distributed around a full circle. A reference marker points north. "Within an angle" means that tolerance on **either side** of north, including wraparound.

(C3) Compare the probability of a heading within $\pi/9$ radians of north with that of a heading within 15° of north. Calculate both probabilities and state which exceeds the other and by how much. **[3]**

6. Charging-space availability [C4-0.5]

Charging spaces at a logistics centre become free according to a constant-rate Poisson process, with a mean waiting time of 4 minutes between openings. Observe from a fixed time.

(C4) Find the probability of waiting more than 6 minutes for the next opening. **[0.5]**

7. Inventory alerts [C4-0.5]

A distributor's stock alerts follow a constant-rate Poisson process averaging 5 alerts every 30 minutes.

(C4) Find the probability that the next alert arrives within 4 minutes of a fixed observation time. **[0.5]**

8. Incoming export orders [C4-1]

An exporter's orders arrive according to a constant-rate Poisson process averaging 3 orders every 20 minutes.

(C4) From a fixed observation time, find the probability that the next order arrives between 2 and 7 minutes later. **[1]**

9. Electronic receipts [C4-1]

Receipts are issued according to a constant-rate Poisson process. The probability that the next receipt is issued within 6 minutes of a fixed observation time is 0.80.

(C4) Infer the rate and calculate the expected number of receipts issued in 15 minutes. **[1]**

10. Wholesale enquiries [C4-1]

Wholesale enquiries reach a cooperative according to a constant-rate Poisson process. From a fixed observation time, the probability of waiting more than 10 minutes for the next enquiry is 0.40.

(C4) Infer the average enquiry rate and express it in enquiries per hour. **[1]**

11. Two payment channels [C4-2]

Completed payments on channels A and B follow independent constant-rate Poisson processes. At a fixed observation time, the probability that A's next payment occurs within 5 minutes is 0.75. The probability that B's next payment takes more than 5 minutes is 0.50. Count payments on both channels from that same observation time.

(C4) Infer both rates. After how many minutes does A's expected payment count exceed B's expected count by 3? Find the time for this difference in expectations. **[2]**

12. Two labelling stations [C5-2]

At a food cooperative, let A and B be labelling times in minutes at two stations. Both have support $[3, 15]$.

For A, equal-length time intervals within the support have equal probabilities. For B, the density rises linearly from zero at 3 to a single peak at 6, then falls linearly to zero at 15.

(a) Identify A's distribution and calculate its mean, variance, and standard deviation. **[1]**

(b) Identify B's distribution and calculate its mean, variance, and standard deviation. **[1]**

13. Custom display production [C5-2]

Two workshops model production time on $[1, 10]$ hours. At I, the density increases along a straight line from zero at 1 to its highest value at 10. At D, it decreases along a straight line from its highest value at 1 to zero at 10. Both densities are zero outside the support.

(a) Identify I's distribution and calculate its mean, variance, and standard deviation. **[1]**

(b) Identify D's distribution and calculate its mean, variance, and standard deviation. **[1]**

14. Requests and daily revenue [C5-2]

A repair cooperative receives requests according to a constant-rate Poisson process averaging 12 requests per hour. Let T be the interarrival time in minutes.

The cooperative models daily revenue R, measured in **thousands of Colombian pesos (COP)**, by an exact normal distribution. Its density is centred at 1800 and has inflection points at 1680 and 1920, all in thousands of COP.

(a) Identify T's distribution and calculate its mean and standard deviation in minutes and its variance in minutes^2. **[1]**

(b) Infer R's parameters. Calculate the mean and standard deviation in **COP** and the variance in **COP**^{2}. Interpret the mean in context. **[1]**

Empirical probability comparisons: C6

For Questions 15–18, calculate absolute error $E = |\widehat{p} - p|$ and percentage error $100E/p$, where $\widehat{p}$ is the empirical probability and p is the stated theoretical probability. Square brackets include an endpoint; parentheses exclude it.

15. Parcel masses, raw data [C6-1]

A sample of 20 parcel masses X, in kilograms, is

1.2	1.5	2.0	2.2	2.4
2.5	2.7	3.0	3.1	3.4
1.8	2.0	2.3	2.6	2.9
3.0	3.2	3.5	3.8	4.0

The model gives $P(2 \le X < 3) = 0.40$; include 2 and exclude 3.

(a) Find the empirical probability of this event. **[0.5]**

(b) Calculate its absolute and percentage errors. **[0.5]**

16. Loading times, frequency data [C6-1]

Recorded loading times X in minutes have frequencies

x_i	2	4	6	8	10	12
f_i	3	5	7	6	5	4

The model gives $P(4 < X \le 10) = 0.65$; exclude 4 and include 10.

(a) Find the empirical probability of this event. **[0.5]**

(b) Calculate its absolute and percentage errors. **[0.5]**

17. File transfers, grouped data [C6-2]

Transfer times X in seconds are grouped below. Classes include the left endpoint and exclude the right endpoint.

Class	$[0, 4)$	$[4, 8)$	$[8, 12)$	$[12, 16)$
f_i	3	9	5	3

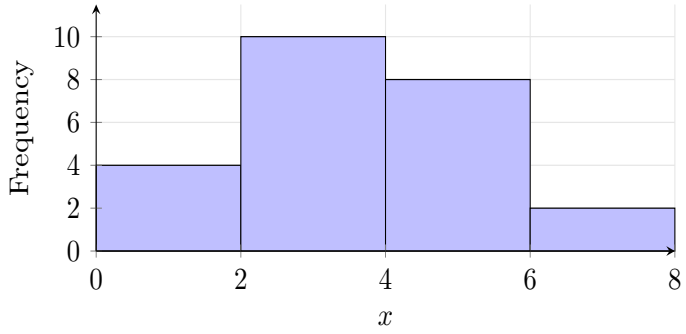
The model gives $P(4 \le X < 12) = 0.75$.

(a) Find the empirical probability of this event. **[1]**

(b) Calculate its absolute and percentage errors. **[1]**

18. Collection queues, histogram [C6-2]

The histogram is the only supplied sample record of collection waiting times X in minutes. Equal-width classes include the left endpoint and exclude the right endpoint. Bar heights are frequencies; the horizontal axis gives minutes.



The model gives $P(2 \le X < 6) = 0.70$.

(a) Find the empirical probability of this event. **[1]**

(b) Calculate its absolute and percentage errors. **[1]**

Sample statistics and errors: C7

For Questions 19–22, calculate the sample mean, sample variance, and sample standard deviation. Always use the variance denominator $n-1$. The theoretical standard deviation is the square root of the stated theoretical variance. For **each of the three statistics**, calculate absolute error $E = |\text{sample statistic} - \text{theoretical statistic}|$ and percentage error $100E/|\text{theoretical statistic}|$. Use unrounded statistics to calculate errors. Round final statistics and absolute errors to 3 decimals and percentage errors to 2 decimals. For grouped data and histograms, explicitly use **class-midpoint estimates**.

19. Cold-room readings, raw data [C7-1]

Five cold-room readings are $6, 8, 9, 11, 12$ degrees Celsius. The theoretical mean is $9\,^\circ\text{C}$ and variance is $5\,(^\circ\text{C})^2$.

(a) Calculate the three sample statistics. **[0.5]**

(b) Calculate absolute and percentage errors for all three statistics. **[0.5]**

20. Sales counts, frequency data [C7-1]

Units sold in seven short sales periods are summarised below. The theoretical mean is 4 units and variance is 3 units^2.

x_i	2	4	6
f_i	2	3	2

(a) Calculate the three sample statistics. **[0.5]**

(b) Calculate absolute and percentage errors for all three statistics. **[0.5]**

21. Courier routes, grouped data [C7-2]

Route times in minutes are grouped below. Classes include the left endpoint and exclude the right endpoint. The theoretical mean is 14 minutes and variance is 12 minutes^2.

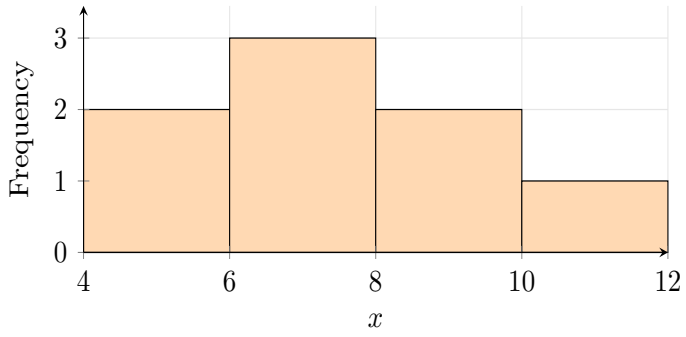
Class	$[8, 12)$	$[12, 16)$	$[16, 20)$
f_i	2	3	1

(a) Use class midpoints to estimate the three sample statistics. **[1]**

(b) Calculate absolute and percentage errors for all three estimates. **[1]**

22. Reusable-container masses [C7-2]

The histogram is the only supplied sample record of container masses in kilograms. Equal-width classes include the left endpoint and exclude the right endpoint. Bar heights are frequencies; the horizontal axis gives kilograms. The theoretical mean is 7 kg and variance is 4 kg^2.



(a) Use class midpoints to estimate the three sample statistics. **[1]**

(b) Calculate absolute and percentage errors for all three estimates. **[1]**